Table 4. Comparison of Fusion Strategies

Fusion strategy	Component models	OS		PFS	
		C-index	*P*	C-index	*P*
A. Level of fusion					
Early fusion	[C + R + I]	0.688 ± 0.054	0.100	0.655 ± 0.048	0.661
Late fusion, three-way	C, R, I	0.705 ± 0.039	0.036	0.655 ± 0.028	0.164
Late fusion, two-way (proposed)	[C + R], I	0.722 ± 0.033	Reference	0.647 ± 0.029	Reference
B. Grouping of modalities in two-way late fusion					
Late fusion, two-way	[C + I], R	0.689 ± 0.034	0.010	0.645 ± 0.041	0.875
Late fusion, two-way	[R + I], C	0.673 ± 0.047	0.046	0.641 ± 0.038	0.788
C. Adaptive fusion					
Attention fusion[a]	[C + R + I]	0.712 ± 0.037	0.525	0.648 ± 0.062	0.978
Gated fusion[b]	[C + R], I	0.723 ± 0.033	0.178	0.650 ± 0.026	0.357

Mean ± SD of the C-index across the five test folds (n = 238); *P*, paired *t* test versus the proposed model. Brackets, modalities trained jointly in one model. [a]Modality features combined by self-attention. [b]Cox weights multiplied by patient-specific gates (from the two risk scores); identical to the proposed model when the gate parameters are zero. Inner cross-validation turned the gates practically off in 3 of 5 folds (OS) and 2 of 5 (PFS). C = clinical variables, I = PET-CT image, OS = overall survival, PFS = progression-free survival, R = radiology report, SD = standard deviation